Planning and Reinforcement Learning in AI Systems

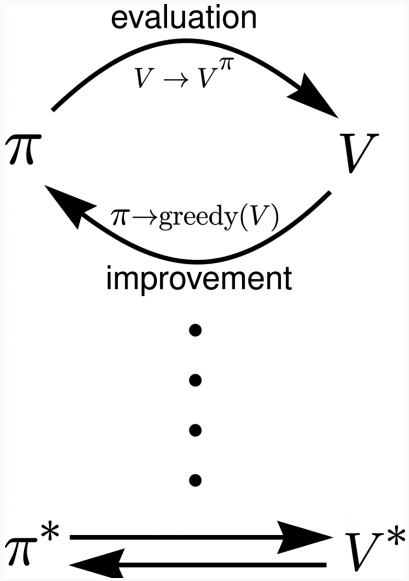
(PARLAI)

Lesson 6a - Model-Free RL (part 1)

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Reminder: The Unified Structure of an RL Algorithm



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Bias and Variance: Two Sources of Estimation Error

In reinforcement learning, we estimate quantities such as

$$V\pi(s), \qquad Q^{\pi}(s,a).$$

Since these estimates are based on finite data, they are subject to two fundamental sources of error.

Bias

Systematic error. How close is the estimator's expected value to the true value?

Variance

Random error. How much would the estimate change if we collected a different set of experiences?

Learning algorithms strive for both low bias and low variance, but reducing one often increases the other.

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Formal Definitions

Suppose $\hat{\theta}$ estimates an unknown quantity θ.

Bias

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

Measures the systematic error of the estimator.

Variance

$$\text{Var}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right].$$

Measures how much the estimate changes across different datasets.

Bias measures accuracy. Variance measures consistency.

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Model Free RL: The Setup

The Setup for Model-Free RL

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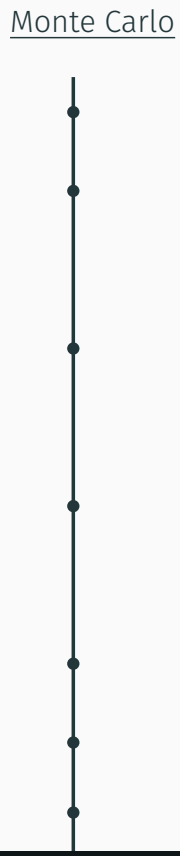
We want to learn a policy that achieves some objective, but can't/don't want to learn the environment dynamics.

We may be able to run many trajectories in the environment or only have access to a simulator that we can run to collect data (we need a lot of it!).

We can adopt several approaches in this setup. We start with Monte-Carlo.

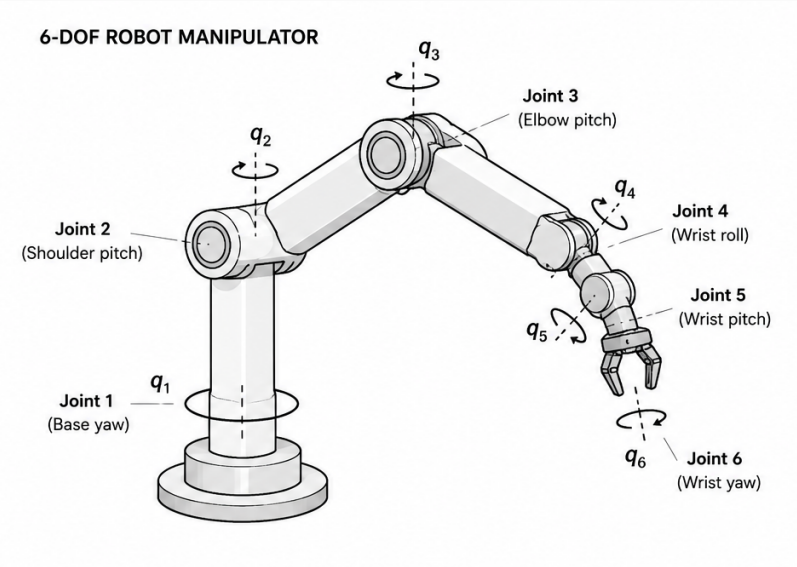
Monte Carlo Simulation-Rollout

Each rollout $\tau = s_0, a_1, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$ involves simulating a sequence of actions, typically by selecting actions according to some policy (e.g., uniformly at random, or based on a heuristic) until a terminal state is reached.



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Model Free RL: The Setup

Monte Carlo Value Estimation

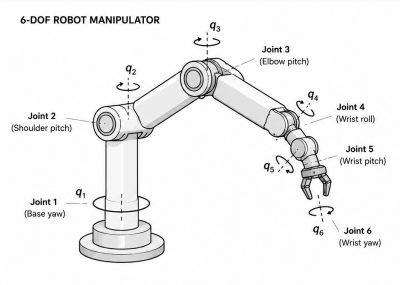
Challenges

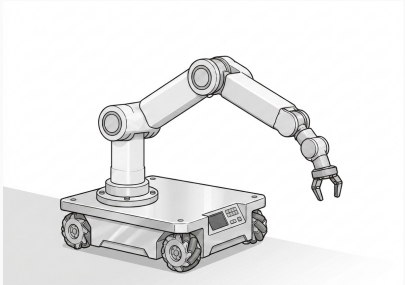
Monte Carlo Control

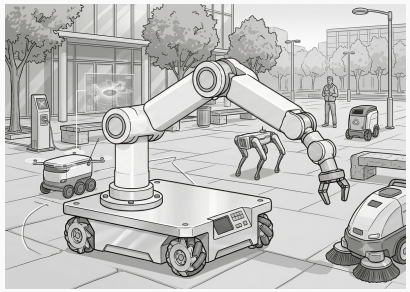
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Monte Carlo Simulation-Rollout

- Rollouts are used to statistically analyze the behavior of complex systems and processes that are influenced by random variables.
- Typically, a large number of simulations are used to estimate the probabilities and distributions of different outcomes.
- Used in different algorithms (such as MCTS)

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Monte Carlo Simulation-Rollout

How does value propagate for a rollout

$$\tau = s_0, a_1, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$$

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Monte Carlo Simulation-Rollout

How does value propagate for a rollout

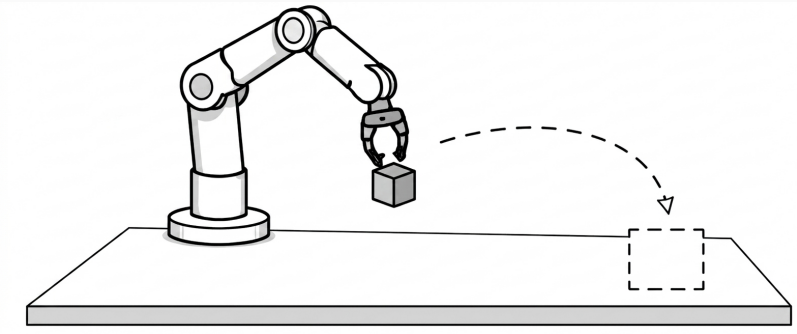
$$\tau = s_0, a_1, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$$

The return for each state is the total [discounted] future reward from each time step: $G_t = r_{t+1} + \gamma G_{t+1}$ with $G_T = 0$.

When is this computed?

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Model Free RL: The Setup

Monte Carlo Value Estimation

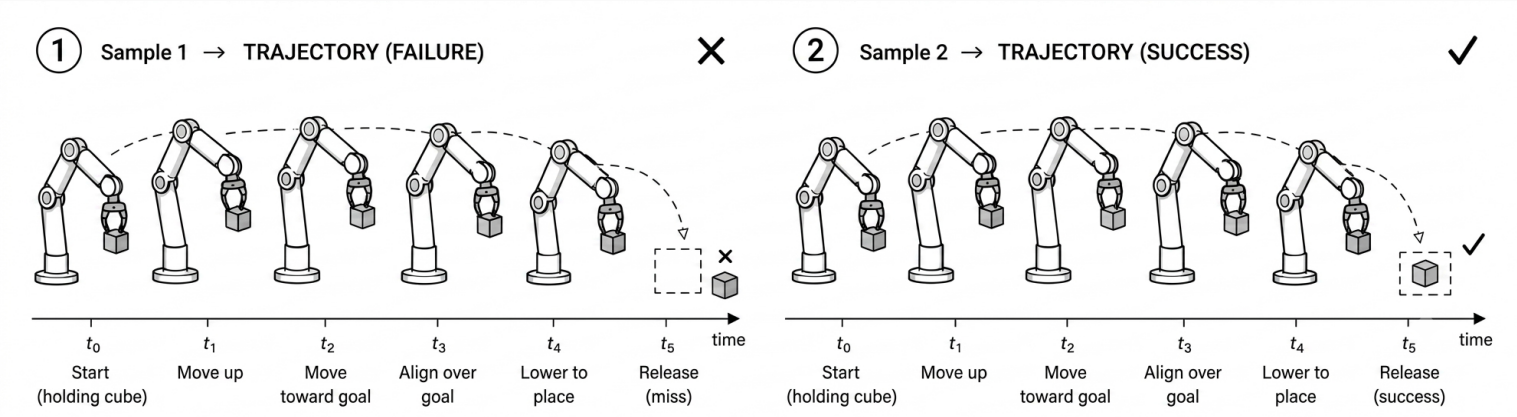
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Monte Carlo Value Estimation

Monte Carlo Value Estimation

Given a policy π, our objective is to estimate the value function

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid s_t = s].$$

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Monte Carlo Value Estimation

Given a policy π, our objective is to estimate the value function

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid s_t = s].$$

Since the true expectation is unknown, we approximate it by averaging sampled returns:

For a state s, the value estimate is

$$\hat{V}^{\pi}(s) = \frac{1}{N(s)} \sum_{i=1}^{N(s)} G^{(i)}(s).$$

where $G_t^{(i)}$ is the return observed for the i-th visit to state s (from the state onward) and $N(s)$ is the number of visits to s

Each rollout provides one sample of the return. By the **Law of Large Numbers**, $\hat{V}_N(s) \longrightarrow V\pi(s)$ as $N \to \infty$.

Anytime: The estimate improves as more rollouts are collected.

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First-Visit vs. Every-Visit Monte Carlo

Consider the trajectory

$$\tau = s_0, a_1, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$$

Key Idea

Use observed reward-to-go of the state as the direct evidence of the actual expected utility of that state.

Two common approaches exist:

- **First-Visit Monte Carlo**
  - Use only the first occurrence of each state within an episode.
- **Every-Visit Monte Carlo**
  - Use every occurrence of the state when computing the average return.

Both estimators converge to the correct value, although their statistical properties differ.

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Example: Estimating a State Value

Suppose the following returns were observed for state s:

Episode	$G(s)$
1	8
2	6
3	10

The Monte Carlo estimate is

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Example: Estimating a State Value

Suppose the following returns were observed for state s:

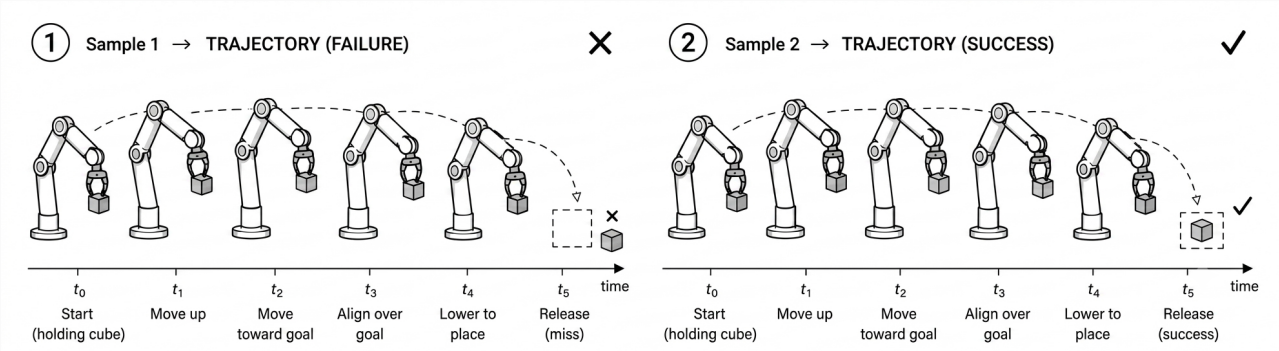
Episode	$G(s)$
1	8
2	6
3	10

The Monte Carlo estimate is

$$\hat{V}(s) = \frac{8 + 6 + 10}{3} = 8.$$

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First-Visit Monte Carlo Policy Evaluation

First-Visit Monte Carlo Policy Evaluation

Initialize:
$\pi \leftarrow$ policy to be evaluated
$V_\pi(s) \leftarrow$ arbitrary value for all $s \in \mathcal{S}$
$\mathit{Returns}(s) \leftarrow \emptyset$ for all $s \in \mathcal{S}$
while true **do**
    Generate an episode following policy π
    for all states s appearing in the episode **do**
        $G \leftarrow$ return following the first occurrence of s
        Append G to $\mathit{Returns}(s)$
        $V_\pi(s) \leftarrow \text{average}(\mathit{Returns}(s))$
    end for
end while

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First-Visit Monte Carlo Policy Evaluation

First-Visit Monte Carlo Policy Evaluation for Q_π

Initialize:
$\pi \leftarrow$ policy to be evaluated
$Q_\pi(s, a) \leftarrow$ arbitrary value for all $s \in \mathcal{S}, a \in \mathcal{A}$
$\mathit{Returns}(s, a) \leftarrow \emptyset$ for all $s \in \mathcal{S}, a \in \mathcal{A}$
while true **do**
    Generate an episode following policy π
    for all state-action pairs (s, a) appearing in the episode **do**
        $G \leftarrow$ return following the first occurrence of (s, a)
        Append G to $\mathit{Returns}(s, a)$
        $Q_\pi(s, a) \leftarrow \text{average}(\mathit{Returns}(s, a))$
    end for
end while

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Challenges with Monte Carlo Evaluation

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Monte Carlo learns from complete sampled episodes.

Monte Carlo methods are simple and unbiased, but they can be slow and noisy.

- Sample inefficiency
- Diminishing returns
- Delayed rewards
- High variance

Sample Inefficiency

Monte Carlo evaluation requires complete episodes before updating value estimates. Thus,

- Each estimate $\hat{V}\pi(s)$ requires sampled rollouts.
- Long episodes delay learning.
- Many episodes may be needed before estimates stabilize.
- Partial trajectories are not immediately used for updates.

This motivates methods that update before the episode ends

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Diminishing Returns

Increasing the number of Monte Carlo rollouts improves the estimate, but each additional rollout contributes a smaller improvement.

The uncertainty of the estimate decreases proportionally to $\frac{1}{\sqrt{N}}$, where N is the number of sampled episodes.

Episodes	Relative Error
100	1.00
400	0.50
1600	0.25
6400	0.125

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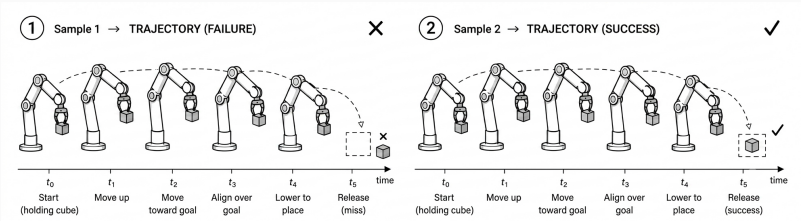
Challenges of Delayed Rewards

Suppose the only reward is received at the end of the episode:

$$r_T = 1, \qquad r_t = 0 \;\; \forall t < T.$$

Two related effects make learning difficult:

- Credit assignment problem
- Discounted learning signal



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Challenges of Delayed Rewards

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Credit Assignment Problem

- A terminal reward provides only indirect feedback.
- It is difficult to determine which earlier actions caused success or failure.
- The longer the episode, the harder this attribution becomes.

Discounted Learning Signal

For a successful episode $G_t = \gamma^{T-t}$

- The reward signal decays exponentially with distance.
- Early states receive only a small learning signal.

Challenges of Delayed Rewards

Discounting weakens the learning signal

$$s_0, r_0 = 0 \to s_1, r_1 = 0 \to s_2, , r_2 = 0 \to \cdots \to s_{T-1}, r_{T-1} = 0 \to s_T, r_T = 1$$

The Monte Carlo return propagates backward:

$$\begin{aligned} G_{T-1} &= \gamma \\ G_{T-2} &= \gamma^2 \\ &\vdots \\ G_0 &= \gamma^T \end{aligned}$$

Distance from reward	1	10	50	100
$\gamma = 0.95$	0.95	0.60	0.077	0.006

The reward reaches every state on the trajetory, but its influence decreases exponentially.

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Variance

Monte Carlo estimates are based on random rollouts.

When used in an uncertain environment, different episodes may produce different returns.

The variance of a random variable X is

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

For Monte Carlo returns,

$$\mathrm{Var}(G_t) = \mathbb{E}\left[(G_t - V\pi(s_t))^2\right].$$

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High Variance in Monte Carlo

Monte Carlo uses the full return:

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots + \gamma^{T-t-1} r_T.$$

This return depends on all future randomness in the episode:

- stochastic actions,
- stochastic transitions,
- stochastic rewards,
- different episode lengths.

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Bias

An estimator is **biased** if its expected value differs from the true quantity.

For an estimator $\hat{\theta}$ of a parameter θ,

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

Applied to value estimation,

$$\text{Bias}(\hat{V}(s)) = \mathbb{E}[\hat{V}(s)] - V\pi(s).$$

Monte Carlo estimation is **unbiased**:

$$\mathbb{E}[\hat{V}(s)] = V\pi(s)$$

But it often has **high variance**

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Monte-Carlo Estimation

Monte Carlo evaluation is attractive because it is simple and unbiased.

However, it has several important challenges:

- It requires complete episodes.
- Long-horizon rewards create difficult credit assignment.
- Discounting weakens distant reward signals.
- Returns can have high variance.
- Reducing error requires many additional samples.

These limitations motivate Temporal Difference learning.

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Monte Carlo Control

Monte-Carlo: Prediction (Evaluation)

First-Visit Monte Carlo Policy Evaluation

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        Append G to $\mathit{Returns}(s)$
        $V_\pi(s) \leftarrow \text{average}(\mathit{Returns}(s))$
    end for
end while

How can we use this for control ?

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Monte-Carlo Control

Monte Carlo Control with Exploring Starts

Initialize:
$Q(s, a) \leftarrow$ arbitrary value for all $s \in \mathcal{S},\ a \in \mathcal{A}(s)$
$\pi(s) \leftarrow$ arbitrary action for all $s \in \mathcal{S}$
Returns$(s, a) \leftarrow \emptyset$ for all $s \in \mathcal{S},\ a \in \mathcal{A}(s)$
while true **do**
    Choose $S_0 \in \mathcal{S}$ and $A_0 \in \mathcal{A}(S_0)$ such that every state–action pair has nonzero probability of being selected
    Generate an episode starting from (S_0, A_0), following policy π
    for all state–action pairs (s, a) appearing in the episode **do**
        $G \leftarrow$ return following the first occurrence of (s, a)
        Append G to *Returns*(s, a)
        $Q(s, a) \leftarrow \text{average}(\textit{Returns}(s, a))$
    end for
    for all states s appearing in the episode **do**
        $\pi(s) \leftarrow \arg\max_a Q(s, a)$
    end for
end while

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Monte Carlo Control with Exploring Starts

Initialize:
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    end for
    for all states s appearing in the episode **do**
        $\pi(s) \leftarrow \arg\max_a Q(s, a)$
    end for
end while

Problem with this approach?

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Monte-Carlo: From Prediction (Evaluation) to Control

- Use a random policy to simulate many (!) trajectories
- Compute the q-value of each state-action pair
- Update π by taking the max action.

Problem with this approach?

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Monte-Carlo Policy Control - Taxi

```
+---------+
|R: | : :G|
| : | : : |
| : : : : |
| | : | : |
|Y| : |B: |
+---------+
  (North)
num episodes completed:   1
total rewards:            -133
mean rewards per episode: -133.00
```

```python
def build_decision_dict(raw_data):
    # Nested dictionary for: State -> Action -> Reward List
    state_action_scores = defaultdict(lambda: defaultdict(lambda: []))
    for trajectory in raw_data:
        reward_sum = 0
        # iterate backwards to calculate the return G of each observed state action pair
        for state, action, reward in reversed(list(zip(trajectory.observations, trajectory.actions, trajectory.rewards))):
            reward_sum += reward
            state_action_scores[state][action].append(reward_sum)

    for state, action_values in state_action_scores.items():
        for action, values_list in action_values.items():
            # Calculate the mean of all returns for a state action pair
            state_action_scores[state][action] = np.mean(values_list)
        # For each state choose the action with the highest mean return
        state_action_scores[state] = max(state_action_scores[state], key=state_action_scores[state].get)
    return state_action_scores
```

https://github.com/CLAIR-LAB-TECHNION/FSTMA-course/blob/main/tutorials/tut03/Monte_Carlo.ipynb

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Monte-Carlo Control - Taxi

```
env.seed(seed)

policy = calc_final_policy(RandomTraveler, 100, "mcc_100")
evaluate_and_print(policy)
policy = calc_final_policy(RandomTraveler, 1000, "mcc_1000")
evaluate_and_print(policy)
policy = calc_final_policy(RandomTraveler, 10000, "mcc_10000")
evaluate_and_print(policy)
policy = calc_final_policy(RandomTraveler, 100000, "mcc_100000")
evaluate_and_print(policy)
```

```
100% 10000/10000 [00:01<00:00, 5139.08it/s]
Monte Carlo Control Policy
-----------------
total reward over all episodes: -1306708
mean reward per episode:        -130.6708
100% 10000/10000 [00:01<00:00, 5334.19it/s]
Monte Carlo Control Policy
-----------------
total reward over all episodes: -1242335
mean reward per episode:        -124.2335
100% 10000/10000 [00:01<00:00, 6415.93it/s]
Monte Carlo Control Policy
-----------------
total reward over all episodes: -729155
mean reward per episode:        -72.9155
100% 10000/10000 [00:01<00:00, 7840.29it/s]
Monte Carlo Control Policy
-----------------
total reward over all episodes: -154428
mean reward per episode:        -15.4428
```

Limitations?

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Monte Carlo Value Estimation

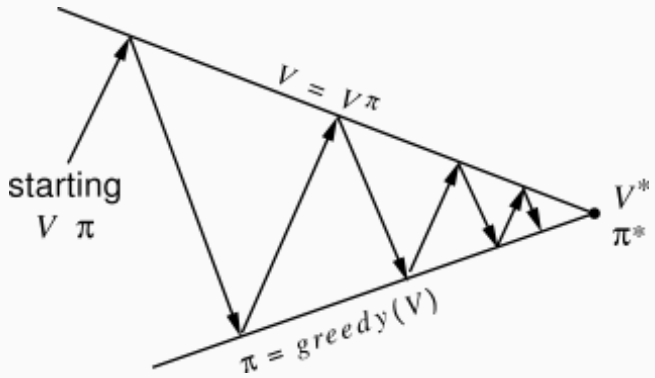
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Generalised Policy Iteration With Monte-Carlo Evaluation



Policy evaluation: Monte-Carlo policy evaluation, $V = V_\pi$?

Policy improvement: Greedy policy improvement?

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Example of Greedy Action Selection

There are two doors in front of you:
You open the left door and get reward 0
$V(left) = 0$
You open the right door and get reward +1
$V(right) = +1$
You open the right door and get reward +2
$V(right) = +2$
You open the right door and get reward +2
$V(right) = +2$
.
.
.
Are you sure you've chosen the best door?



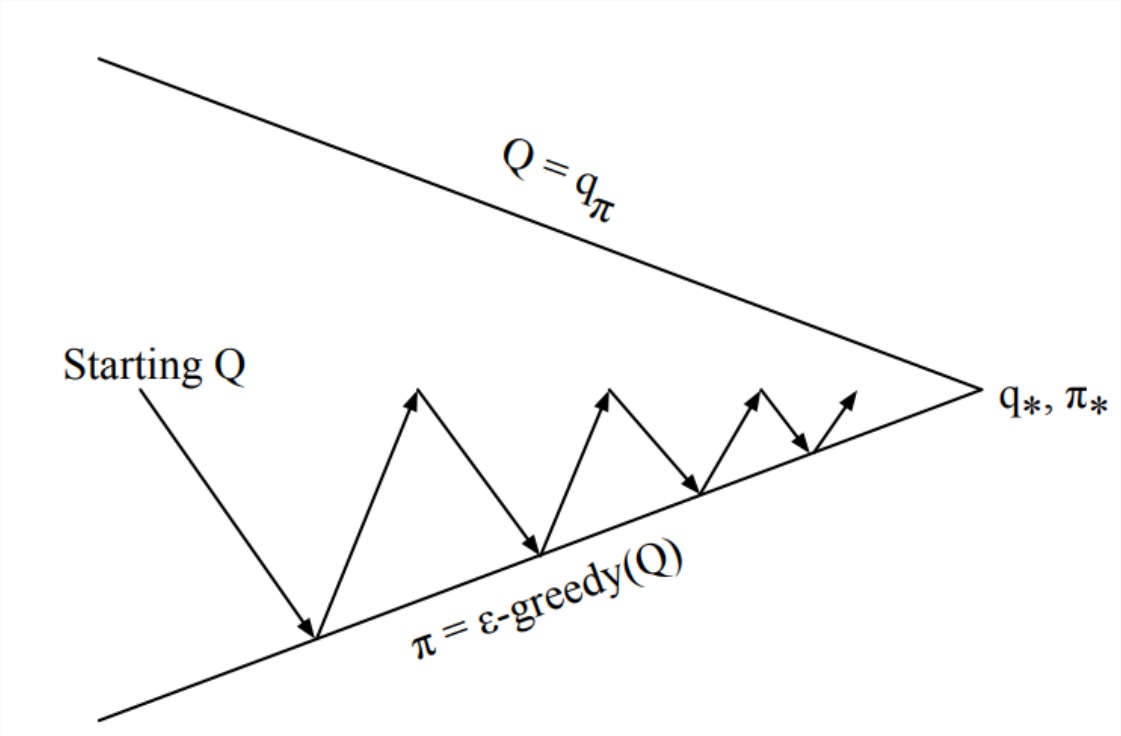
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ϵ-Greedy Exploration

- Simplest idea for ensuring continual exploration
- All m actions are tried with non-zero probability
- With probability $1 - \epsilon$ choose the greedy action
- With probability ϵ choose an action at random

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Monte-Carlo Control

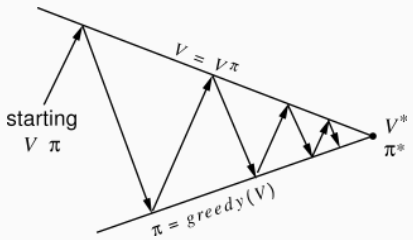


Every Iteration:

- **Policy evaluation:** Monte-Carlo policy evaluation. $Q \approx q_\pi$
- **Policy improvement:** ϵ-greedy improvement

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Monte-Carlo Policy Iteration



- **Policy evaluation:** Monte-Carlo policy evaluation. $Q_\pi(s, a)$
- Policy improvement: ϵ-greedy improvement

Is it necessary to wait until the end of execution of all trajectories ?

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Incremental Mean

The mean $\mu_1, \mu_2, \ldots$ of a sequence $x_1, x_2, \ldots$ can be computed incrementally,

$$\mu_k = \frac{1}{k} \sum_{j=1}^{k} x_j$$

$$= \frac{1}{k} \left(x_k + \sum_{j=1}^{k-1} x_j \right)$$

$$= \frac{1}{k} \left(x_k + (k-1)\mu_{k-1} \right)$$

$$= \mu_{k-1} + \frac{1}{k}(x_k - \mu_{k-1})$$

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Incremental Monte-Carlo Updates

- Update $V(s)$ incrementally after episode $S_1, A_1, R_2, ..., S_T$
- For each state S_t with return G_t:

$$N(S_t) \leftarrow N(S_t) + 1$$

$$V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)}(G_t - V(S_t))$$

- In non-stationary problems, it can be useful to track a running mean, i.e. forget old episodes.

$$V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$$

where α is the [decaying] learning rate

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Monte-Carlo Control

On-Policy First-Visit Monte Carlo Control (ϵ-soft)

Algorithm parameter: $\epsilon > 0$
Initialize:
$\pi \leftarrow$ arbitrary ϵ-soft policy
$Q(s,a) \leftarrow$ arbitrary, for all $s \in \mathcal{S},\ a \in \mathcal{A}(s)$
$\mathit{Returns}(s,a) \leftarrow \emptyset$, for all $s \in \mathcal{S},\ a \in \mathcal{A}(s)$
while true **do**
    Generate an episode following π:
    $s_0, a_0, r_1, \ldots, s_{T-1}, a_{T-1}, r_T$
    $G \leftarrow 0$
    for $t = T-1, T-2, \ldots, 0$ **do**
        $G \leftarrow \gamma G + r_{t+1}$
        if (s_t, a_t) is the first occurrence in the episode **then**
            Append G to $\mathit{Returns}(s_t, a_t)$
            $Q(s_t, a_t) \leftarrow$ average($\mathit{Returns}(s_t, a_t)$)
            $a^* \leftarrow \arg\max_a Q(s_t, a)$
            for all $a \in \mathcal{A}(s_t)$ **do**
                if $a = a^*$ **then**
                $\pi(a|s_t) \leftarrow 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s_t)|}$
                else
                $\pi(a|s_t) \leftarrow \frac{\epsilon}{|\mathcal{A}(s_t)|}$
                end if
            end for
        end if
    end for
end while

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Policy Improvement Theorem

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Definition

For policy π and nondeterministic policy $\pi' : \mathcal{S} \times \mathcal{A} \to [0, 1]$, define:

$$Q_\pi(s, \pi'(s)) = \sum_{a \in \mathcal{A}} \pi'(a|s) Q_\pi(s, a)$$

Theorem

Let π and π' be any pair of policies (deterministic or nondeterministic). If for all $s \in \mathcal{S}$ $Q_\pi(s, \pi'(s)) \geq V_\pi(s)$ then for all $s \in \mathcal{S}$, $V_{\pi'}(s) \geq V_\pi(s)$. i.e., π' is an improvement over π.

π', that can either exploit or explore, must be at least as good as π

ϵ-Greedy Policy Improvement

Definition

A nondeterministic policy π is **ϵ-soft** if
$\forall s \in \mathcal{S}, a \in \mathcal{A} \quad \pi(|s) \geq \frac{\epsilon}{|\mathcal{A}|}$

Definition

A nondeterministic policy π is called **ϵ-greedy** with respect to Q if for some action $a^* \in \arg\max_{a \in \mathcal{A}} Q(s, a)$

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}|} & a = a^* \\ \frac{\epsilon}{|\mathcal{A}|} & \text{otherwise} \end{cases}$$

Theorem

For any ϵ-soft policy π, if the ϵ-greedy policy π' with respect to Q_π, then $V_{\pi'}(s) \geq V_\pi(s)$.

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ϵ-Greedy Policy Improvement

Theorem

For any ϵ-soft policy π, if the ϵ-greedy policy π' with respect to Q_π, then $V_{\pi'}(s) \geq V_\pi(s)$.

$$Q_\pi(s, \pi'(s)) = \sum_{a \in \mathcal{A}} \pi'(a|s) Q_\pi(s, a) \qquad \text{definition}$$

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$$Q_\pi(s, \pi'(s)) = \sum_{a \in \mathcal{A}} \pi'(a|s) Q_\pi(s, a) \qquad \text{definition}$$

$$= \frac{\epsilon}{m} \sum_{a \in \mathcal{A}} Q_\pi(s, a) + (1 - \epsilon) \max_{a \in \mathcal{A}} Q_\pi(s, a) \qquad \text{definition}$$

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$$= \frac{\epsilon}{m} \sum_{a \in \mathcal{A}} Q_\pi(s, a) + (1 - \epsilon) \max_{a \in \mathcal{A}} Q_\pi(s, a) \qquad \text{definition}$$

Since π is ϵ-soft, then $\frac{\pi(a|s) - \frac{\epsilon}{|\mathcal{A}|}}{1-\epsilon}$ is non-negative and sums to 1.

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For any ϵ-soft policy π, if the ϵ-greedy policy π' with respect to Q_π, then $V_{\pi'}(s) \geq V_\pi(s)$.

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Since π is ϵ-soft, then $\frac{\pi(a|s) - \frac{\epsilon}{|\mathcal{A}|}}{1-\epsilon}$ is non-negative and sums to 1.
Since $\max Q_\pi$ is greater than any weighted sum of Q_π, then

$$\geq \frac{\epsilon}{m} \sum_{a \in \mathcal{A}} Q_\pi(s, a) + (1 - \epsilon) \sum_{a \in \mathcal{A}} \frac{\pi(a|s) - \frac{\epsilon}{m}}{1 - \epsilon} Q_\pi(s, a)$$

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ϵ-Greedy Policy Improvement

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For any ϵ-soft policy π, if the ϵ-greedy policy π' with respect to Q_π, then $V_{\pi'}(s) \geq V_\pi(s)$.

$$Q_\pi(s, \pi'(s)) = \sum_{a \in \mathcal{A}} \pi'(a|s) Q_\pi(s, a) \qquad \text{definition}$$

$$= \frac{\epsilon}{m} \sum_{a \in \mathcal{A}} Q_\pi(s, a) + (1 - \epsilon) \max_{a \in \mathcal{A}} Q_\pi(s, a) \qquad \text{definition}$$

Since π is ϵ-soft, then $\frac{\pi(a|s) - \frac{\epsilon}{|\mathcal{A}|}}{1-\epsilon}$ is non-negative and sums to 1.
Since $\max Q_\pi$ is greater than any weighted sum of Q_π, then

$$\geq \frac{\epsilon}{m} \sum_{a \in \mathcal{A}} Q_\pi(s, a) + (1 - \epsilon) \sum_{a \in \mathcal{A}} \frac{\pi(a|s) - \frac{\epsilon}{m}}{1 - \epsilon} Q_\pi(s, a)$$

$$= \sum_{a \in \mathcal{A}} \pi(a|s) Q_\pi(s, a) = V_\pi(s) \underset{\text{policy improvement}}{\Longrightarrow} V_{\pi'}(s) \geq V_\pi(s)$$

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How do we make sure exploration eventually stops?

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How do we make sure exploration eventually stops?

- Decay ϵ Over Time: e.g., exponential decay $\epsilon_t = \gamma \cdot \epsilon_{t-1}$ with $\gamma \in (0, 1)$.
- After a sufficient number of iterations (when $Q(s, a)$ has stabilized), switch to a purely greedy policy.

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GLIE

Greedy in the Limit with Infinite Exploration (GLIE)

- All state-action pairs are explored infinitely many times,

$$\lim_{k \to \infty} N_k(s, a) = \infty$$

- The policy converges on a greedy policy,

$$\lim_{k \to \infty} \pi_k(a|s) = \mathbf{1}(a = \arg\max_{a' \in \mathcal{A}} Q_k(s, a'))$$

For example, ϵ-greedy is GLIE if α reduces to zero at $\epsilon_k = \frac{1}{k}$.

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GLIE Monte-Carlo Control

- Sample kth episode using $\pi : \{s_1, A_1, R_2, ..., s_T\} \sim \pi$
- For each state s_t and action A_t in the episode,

$$N(s_t, A_t) \leftarrow N(s_t, A_t) + 1$$

$$Q(s_t, A_t) \leftarrow Q(s_t, A_t) + \frac{1}{N(s_t, A_t)}(G_t - Q(s_t, A_t))$$

- Improve policy based on new action-value function

$$\epsilon \leftarrow \frac{1}{k}$$

$$\pi \leftarrow \epsilon\text{-greedy}(Q)$$

Theorem

GLIE Monte-Carlo control converges to the optimal action-value function, $Q(s, a) \rightarrow Q^*(s, a)$

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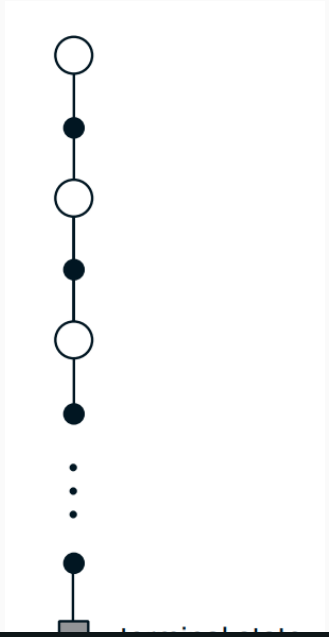
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Conclusion

Monte Carlo (MC)

- MC methods learn directly from episodes of experience
- MC is model-free: no knowledge of MDP transitions / rewards
- MC learns from complete episodes: no bootstrapping.
- MC uses the simplest possible idea: value = mean return
- Caveat: can only apply MC to episodic MDPs: all episodes must terminate



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Monte Carlo (MC)

- Converge very slowly to correct utilities values (requires a lot of sequences)

$$V_\pi(s) = \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

- Doesn't exploit Bellman constraints on policy values

$$V_\pi(s) = R(s) + \gamma \sum_{s'} \mathcal{P}(s'|s, \pi(s) \cdot V_\pi(s')$$

- Can consider estimates that violate this property badly

How can we incorporate such constraints ?

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